TEJAS SATISH NAVALKHE

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Professional Summary

AI/ML and Software Engineer with **over 3 years of experience** building scalable machine learning solutions for fintech, transport, and edtech. I specialise in **Machine Learning**, **Deep Learning**, **NLP** (BERT, GPT), and **MLOps** (Docker, AWS, CI/CD), with a track record of driving impact: delivered a **19.32% revenue boost** via dynamic pricing and cut trading system latency by **40%**. Hold a **Distinction in MSc Data Science (AI Specialisation)** from Newcastle University, UK, and excel in deploying high-impact projects with **Python, TensorFlow, PyTorch, and cloud platforms**.

Work Experience

Deputy Manager (Technology) — **08/2025 – Present**
Jalgaon Janata Sahakari Bank Limited — **Jalgaon, Maharashtra, India**

- **Automated MIS and RBI-format reporting** across 10+ departments using **Python, SQL, FastAPI, Seaborn, and Matplotlib**, reducing manual effort by **70%** and errors by **90%**.
- **Developed centralised data dashboards** for Risk, Loan, IT, and Compliance units, achieving **99% accuracy** and ensuring on-time **RBI submissions**.
- **Built AI/ML predictive models** with **Scikit-learn and Pandas** to enhance **CASA growth and deposit mobilisation**, improving campaign efficiency by **15–20%**.
- **Deployed secure on-premises servers** for automated data storage, retrieval, and report generation, ensuring **RBI and cybersecurity compliance**.
- **Created internal automation tools,** including the **"Link Hub" intranet portal** and **bulk email system**, streamlining workflows and boosting communication efficiency by **80%**.

Freelance AI/ML Engineer — **09/2024 – 06/2025**
Self-Employed — **Remote, India**

- Created **AI/ML solutions for 15+ edtech/fintech clients**, including a real-time trading system that slashed latency by **30%** using Python, Flask, Redis, and WebSocket. Built scalable REST/WebSocket APIs (FastAPI) on AWS (EC2, S3, RDS), cutting deployment time by **25%** while handling **5,000+ concurrent users**.
- Enhanced **NLP and course recommendation engines** (TF-IDF, BERT, GPT) for an edtech platform, lifting user engagement by **25%**. Systematised retraining and CI/CD (GitHub Actions) saved **15 hours/month** and ensured **99.8% uptime**.
- Boosted model accuracy by **12%** via orchestrated hyperparameter tuning (MLflow). Streamlined data pipelines (Apache Airflow) and web scraping accelerated data aggregation by **50%** for real-time predictions.

Data Science Intern — **05/2024 – 09/2024**
Co Wheels Car Club — **Newcastle, United Kingdom**

- Designed a **dynamic pricing model** using ML, analysing data from **30,000+ users** and **600+ vehicles** (including 200+ EVs). Optimised pricing for location, peak hours, and seasonality, driving a **19.32% revenue increase** versus static models.
- Created a **Flask API simulation tool** for real-time pricing adjustments. Automated preprocessing of **40+ features** cut manual work by **~40%** and improved model accuracy.
- Synced data with Co Wheels' booking system, reducing data sync issues by **~25%**. Insights guided their shift to new pricing tool, boosting profits without compromising user satisfaction.

Freelance Software Engineer — **09/2022 – 04/2024**
Finvestox Capital — **Remote, India**

- Architected an **algorithmic trading system** enabling **sub-millisecond execution**, increasing trading volume by **20%** and returns by **10%** in 3 months. Deployed on AWS/Docker for **50% faster scaling**, with Redis caching speeding data retrieval by **35%** at **99.9% uptime**.
- Formulated a **trading platform** with risk management. Selenium-based broker logins cut trade execution by **30%**; Plotly dashboards improved real-time analytics accuracy by **25%**.
- Used **Redis pub/sub** to streamline multi-server data flow and refined SQL databases, cutting prediction latency by **20%** during high-frequency trading.

Web Development Intern — **07/2022 – 09/2022**
Vitaecura Consultancy Services Pvt. Ltd. — **Indore, India**

- Revamped **5+ client-facing web features** (JavaScript, jQuery), improving page load by **20%** and engagement by **15%** via responsive design.

- Streamlined IT workflows to save **10 hours/month** in manual effort while supporting delivery of **3+ IT projects** using Agile.
- **Drove business development** by creating client demo dashboards that secured **2 new clients**, and led UX research, improving site navigation to boost user satisfaction by **10%**.

Education

Master of Science in Data Science (With Specialisation in Artificial Intelligence) **09/2023 – 11/2024**
Newcastle University, Newcastle Upon Tyne, United Kingdom **Grade: Distinction**
Modules: Machine Learning, Deep Learning, Big Data Technology, Image Informatics, Statistics, EDA, Data Management.

Bachelor of Technology in Computer Science Engineering **08/2019 – 07/2023**
Prestige Institute of Engineering Management & Research, Indore, India **CGPA (8.78/10)**
Modules: Data Structure, Cloud Computing, DBMS, Software Engineering, ML, Project Management

Projects

Scalable AI-Driven Algorithmic Trading System with Real-Time Automatic Trading

- Constructed a **low-latency trading system** using Apache Spark Streaming and WebSockets, reducing data delay by **40%** and integrating AI models (LSTM, ARIMA) to improve trade accuracy by **15%** via real-time sentiment analysis.
- Containerised deployment with **Docker on AWS**, achieving **99.9% uptime**, and designed a custom risk-management module to limit losses to **<2% per trade** while optimizing returns.
- Reverse-engineered **AngelOne/Zerodha APIs** to unify order execution across brokers and implemented a Selenium-based testing framework, cutting execution bugs by **30%** and ensuring compliance with trading regu lations.

AI-Driven Face Recognition Attendance System

- Developed a **face recognition-based attendance system** using **OpenCV**, **automating attendance tracking** and reducing manual effort by 70%.
- Achieved an impressive **99.38% recognition accuracy**, ensuring **reliable** and **consistent identification** under varied real-world conditions.
- Integrated **Amazon S3** for **large-scale image storage**, enabling **scalable**, **cloud-ready deployment** with **fast and efficient data access** for real-time face matching.

Technical Skills

Programming Languages: Python, R, JavaScript
Machine Learning & Deep Learning: Scikit-learn, TensorFlow, Keras, PyTorch, XGBoost
Natural Language Processing: spaCy, NLTK, Transformers, Hugging Face, BERT, GPT, LLaMA, LangChain, Haystack, OpenAI APIs
MLOps: MLflow, Weights & Biases (W&B), DVC, Airflow, Docker, Kubernetes
Cloud & DevOps: AWS (EC2, S3, RDS, Lambda), Azure (basic), CI/CD (GitHub Action, Jenkins), Monitoring & Logging
Software Development & Testing: Clean Code Principles, Design Patterns, Unit Testing (Pytest, unittest), Integration Testing, Selenium, A/B Testing, Performance Testing
Version Control & Collaboration: Git, GitHub, Jira
Data & Databases: SQL, PostgreSQL, MongoDB, Redis
Web Technologies & API: RESTful API, WebSocket, FastAPI, Flask, Django, Frontend
Data Engineering & Processing: Pandas, NumPy, Apache Spark, ETL Pipelines
Visualisation: Matplotlib, Seaborn, Plotly, Power BI
Operating Systems: Linux, Unix, Windows, MacOS

Certifications

Machine Learning with Scikit-Learn Madecraft | 08/2024
AWS Educate Getting Started with Compute Amazon Web Services (AWS) | 02/2023
Data Science & Advance Machine Learning using Python Grras Solutions | 01/2022

Awards

- **NCL+ Award** by Newcastle University 05/2024
- **Best Student of the Batch (2019-2023) Award** by Prestige Institute of Engineering Management & Research 04/2023

Publications

- ***'Mathability Level 1: A Mathematics Foundation Course'***, Amazon.in
- ***'Design and Development of Smart Virtual Assistant Using Latest Tools and Technologies'***, International Journal of Core Engineering & Management